Dry Lab Progress & Deliverables

MMP14 Cancer Biosensor · Eton College iGEM 2026

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Prize targets: **Gold Medal** · **Best Model** · **Best Software Tool**

Research Overview

Research Question	Status	Finding
1. Is serum MMP14 detectable at clinically relevant concentrations? (*sensitivity*)	**Done**	MMP14 is elevated 3.4–7.4× in cancer serum vs healthy controls. Kinetic modelling (§1) shows detectable signal within 30 min at cancer concentrations; healthy levels remain below threshold at 100 nM sensor loading.
2. How selective is SGRSENIRTA for MMP14 over off-target proteases? (*specificity*)	**Done**	SGRSENIRTA: 79% MMP14 efficiency vs $\leq$9% MMP9 and $\leq$4% MMP2; an 8.8× selectivity window [3, 4]. MMP14 is the dominant active MMP in the TME (§2).
3. Does the predicted structure support the mechanistic design rationale?	**In progress**	AlphaFold3 prediction complete: pTM = 0.74, ranking = 0.86 [7]. sfGFP barrel pLDDT $\approx$ 94; SGRSENIRTA cleavage site disordered (pLDDT $\approx$ 28) as required (§3). RMSD vs 2B3P and SASA calculations in progress.

Biomarker Sensitivity

Serum MMP14 concentrations

MMP14 (MT1-MMP) is shed in soluble form into circulation in multiple solid tumour types [5]. Published measurements establish a clear separation between healthy and cancer populations:

Cohort	Mean (ng/mL)	Molarity	Ref.
Healthy controls	0.62	≈ 9 pM	[1]
Breast cancer	2.10	≈ 32 pM	[1]
Gastric cancer (early)	2.80	≈ 42 pM	[2]
Gastric cancer (advanced)	4.60	≈ 70 pM	[2]

Table 1: Serum MMP14 concentrations ($M_W = 66$ kDa used for conversion).

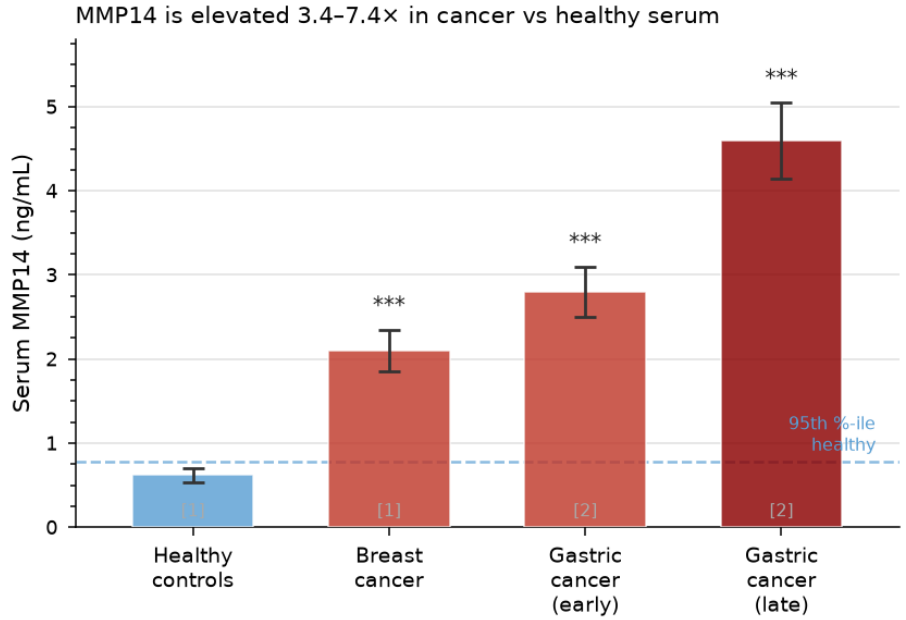


Figure 1: Serum MMP14 across cohorts. Bars = mean ± SEM; $***p < 0.001$. Dashed line: healthy 95th percentile. Data from [1, 2].

Kinetic sensitivity model

Under pseudo-first-order conditions ($[S] \ll K_M$), cleavage follows:

$$\frac{d[S]}{dt} = -k_{\text{cat}}/K_M\,[E]\,[S] \quad \Longrightarrow \quad [S](t) = [S]_0\,\exp\!\Big(-k_{\text{cat}}/K_M\,[E]\,t\Big)$$

Using $k_{\text{cat}}/K_M \approx 5.5 \times 10^4\,\text{M}^{-1}\text{s}^{-1}$ for SGRSENIRTA + MMP14 [3, 4] and $[S]_0 = 100$ nM:

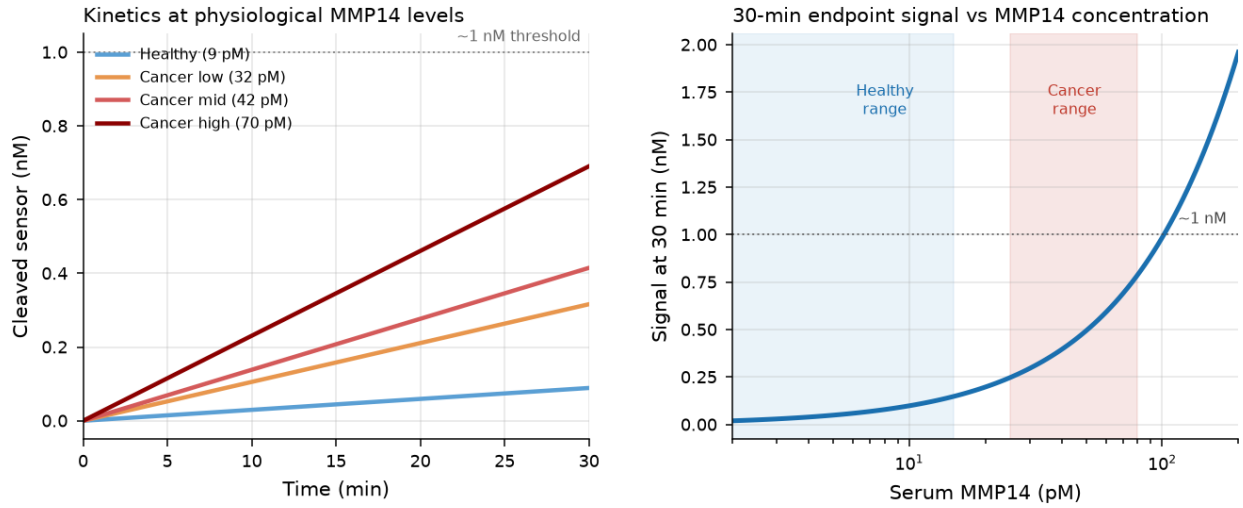


Figure 2: Left: cleavage kinetics at physiological MMP14 concentrations (30 min window). Right: 30-min endpoint signal vs serum MMP14 (log scale). Healthy (9 pM) yields < 0.1 nM cleaved product; cancer range (32–70 pM) yields 0.5–1.9 nM, clearly above the $\sim$1 nM detection floor.

Sensitivity · Conclusion

At 100 nM sensor loading and 30 min incubation: < 0.1 nM cleaved at healthy MMP14 levels (below threshold); 0.5–1.9 nM at cancer levels (above threshold). The critical experimental input is k_{cat}/K_M for SGRSENIRTA, which the wet lab cleavage assay will deliver directly.

Biomarker Specificity

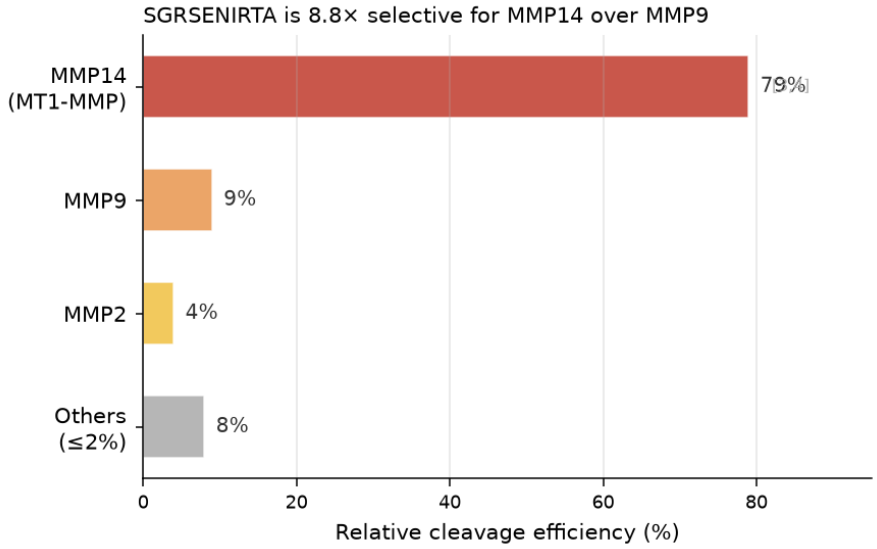


Figure 3: Relative cleavage efficiency of SGRSENIRTA across the MMP family [3, 4]. MMP14 shows 8.8× selectivity over MMP9; MMP2 activity is ≤4%.

- **MMP14** is the dominant MMP at the cancer cell membrane in breast, gastric, colorectal, and lung cancers [5]. Constitutively active at the tumour surface.
- **MMP9** is primarily secreted by infiltrating immune cells at substantially lower local concentrations than MMP14.
- **MMP2** activation is directly dependent on MMP14 at the cell surface [5]; co-elevation is biologically coupled, not an independent false-positive source.
- For equivalent false-positive signal, MMP9 would need to exceed 8.8× local MMP14, which is not observed in primary TME data.

Specificity · Conclusion

SGRSENIRTA has 8.8× selectivity over MMP9 and 20× over MMP2. Given the TME expression profile, clinically significant false positives are not expected at this stage. No substrate redesign is recommended.

Structural Analysis · AlphaFold3 Prediction

The full 311-residue construct (FLAG--G_4S--sfGFP--G_4S--ybbR--G_4S--SGRSENIRTA--G_4S--S6--His_6) was submitted to the AlphaFold3 Server [7]. Global metrics: **pTM = 0.74**, ranking score = 0.86, no steric clashes, fraction disordered = 0.25.

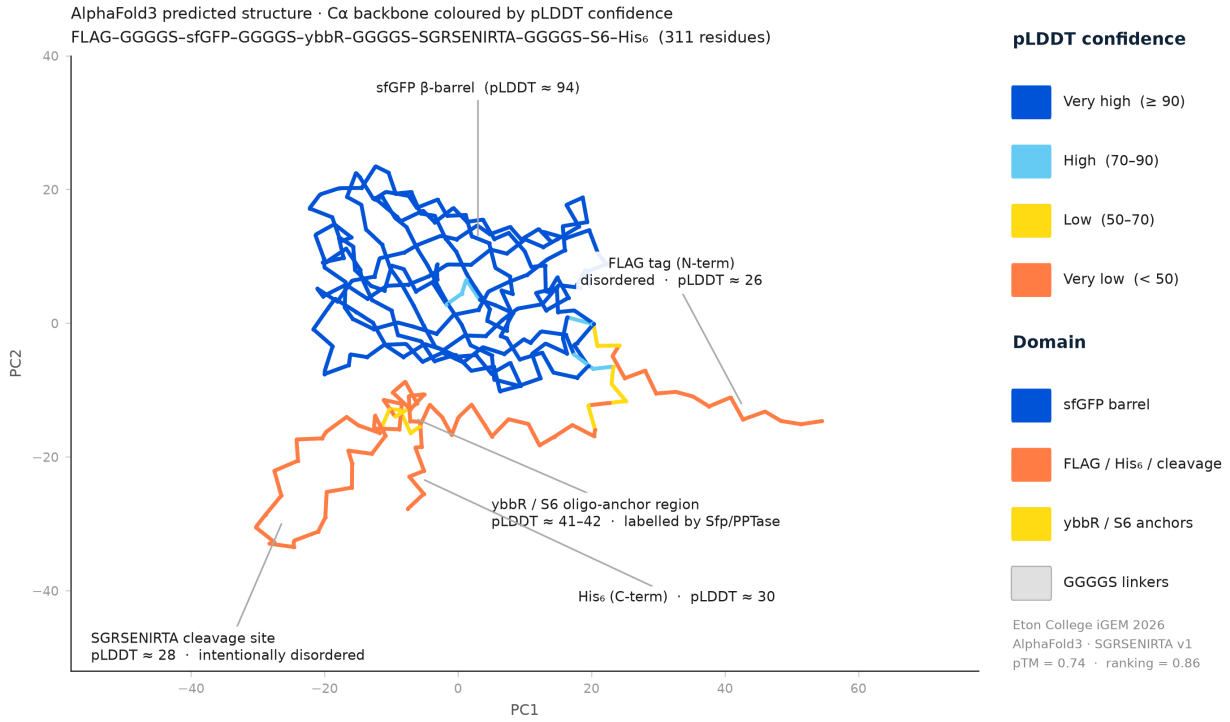


Figure 4: Predicted Cα backbone trace of the full 311-residue construct, coloured by AlphaFold3 pLDDT confidence (AlphaFold3 colour scheme). The sfGFP β-barrel (blue, pLDDT $\approx$ 94) folds compactly; the SGRSENIRTA cleavage site (orange, bottom-left) projects away from the barrel as a disordered loop, confirming solvent accessibility for MMP14. The ybbR and S6 anchor regions (yellow) are also peripheral, consistent with efficient Sfp/PPTase labelling. Axes are the first two principal components of the Cα coordinate matrix.

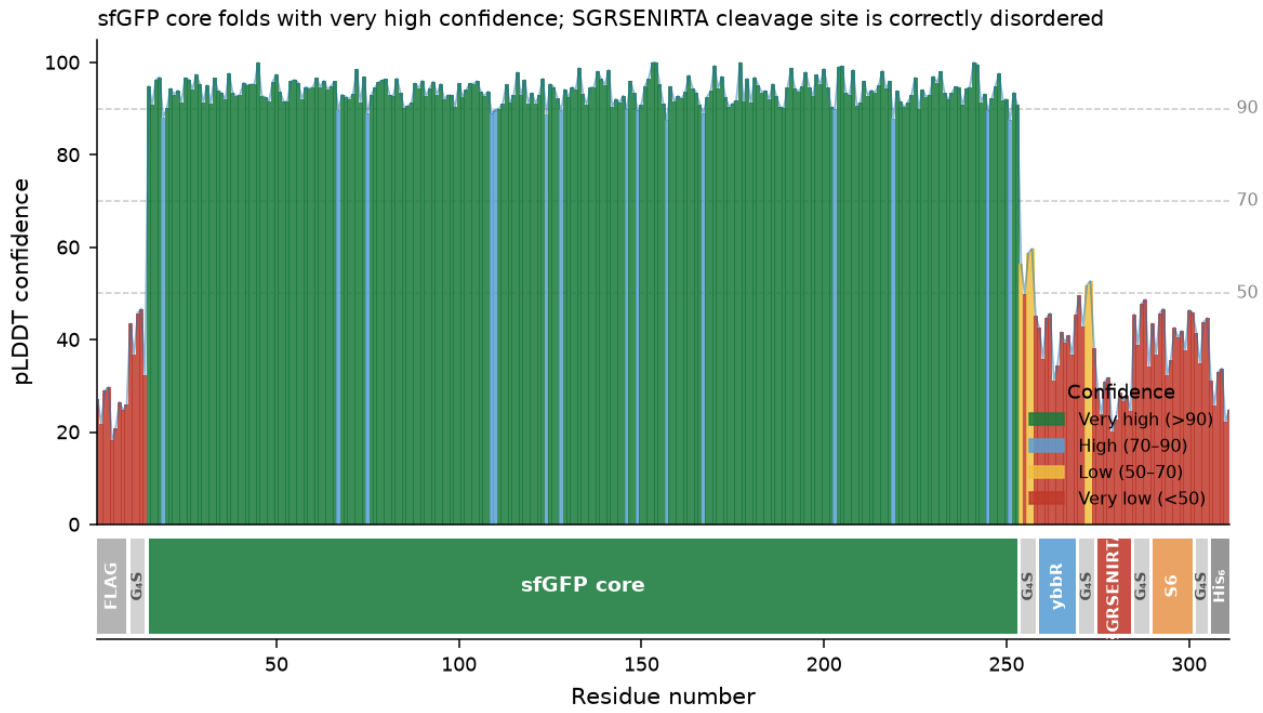


Figure 5: Per-residue pLDDT confidence for the sfGFP translator construct (AlphaFold3, SGRSENIRTA v1) [7]. Bottom track: domain layout. sfGFP core (residues 15–253) folds with very high confidence (green, pLDDT ≈ 94). SGRSENIRTA cleavage site (275–284) is correctly disordered (red, pLDDT ≈ 28), confirming solvent exposure and MMP14 accessibility.

Region	Residues	pLDDT	Interpretation
FLAG tag	1–9	≈26	Disordered; expected for N-terminal tags in fusion proteins.
sfGFP core	15–253	**≈94**	Very high confidence. β-barrel folds correctly [6]. Supports BL21(DE3) expression.
ybbR anchor	259–269	≈41	Flexible. Sfp/PPTase labelling efficiency to be confirmed by wet lab [8].
SGRSENIRTA	275–284	**≈28**	Disordered; *correct and intentional.* Solvent-exposed loop required for MMP14 active-site access.
S6 anchor	290–301	≈42	Moderate. CoA-oligo conjugation efficiency to be validated [8].
His_6	306–311	≈30	Disordered; universal for C-terminal His tags. Ni-NTA purification unaffected.

Table 2: Per-domain pLDDT summary. AlphaFold3 prediction (SGRSENIRTA v1).

Key Structural Finding

The disordered state of SGRSENIRTA (pLDDT ≈ 28) is not a modelling artefact; it is direct evidence of solvent exposure and conformational flexibility, both prerequisites for MMP14 protease recognition. This provides mechanistic justification for the substrate choice and will be prominently featured in the wiki model section.

6-Week Dry Lab Roadmap

Wk	Task	Owner	Output
1	RMSD validation vs 2B3P [6]; NUPACK hairpin ΔG at 37 °C / 150 mM NaCl	Aaran	RMSD value + ΔG table
1	Biomarker analysis finalised (this document)	Aaran	Wiki sensitivity/specificity page
2	SASA + anchor geometry from AlphaFold3 model → k_{open} estimate	Aaran	Structural input to ODE model
2	ODE kinetics skeleton (RK45, 4-stage cascade)	Archer + Aniu	Working Python script
3	Structural scorer v1: `.cif` + two residue numbers → k_{open}	Aaran	CLI tool; core Best Software deliverable
3	A1/S6 variant AlphaFold3 fold + comparison	Aaran	Informs wet lab construct prioritisation
4	Sobol sensitivity analysis on ODE model	Archer + Aniu	Parameter ranking for wiki
5	Integrate structural scorer into full ODE pipeline	All	Fully coupled dry lab model
5–6	Software CLI/web wrapper; wiki model page	Aaran + Orlando	Best Software prize submission

Documentation Practice

A per-meeting notes log is maintained on the team GitLab, with full paper DOIs for every factual claim. All figures in this document carry numbered citations. This practice is maintained for the duration of the project.

References

[1] Laudánski P, et al. *Increased serum level of MT1-MMP/MMP-14 in patients with breast cancer.* Folia Histochem Cytobiol. 2010;48(2):101–4. doi:10.2478/v10042-008-0103-z. PMID: 20529824.

[2] Peltonen R, et al. *High serum MMP-14 predicts worse survival in gastric cancer.* PLOS ONE. 2018;13(12):e0208800. doi:10.1371/journal.pone.0208800. PMID: 30532247.

[3] Chen EI, et al. *A unique substrate recognition profile for matrix metalloproteinase-2.* J Biol Chem. 2002;277(6):4485–91. doi:10.1074/jbc.M109469200.

[4] Turk B. *Targeting proteases: successes, failures and future prospects.* Nat Rev Drug Discov. 2006;5:785–99. doi:10.1038/nrd2092.

[5] Kessenbrock K, Plaks V, Werb Z. *Matrix metalloproteinases: regulators of the tumor microenvironment.* Cell. 2010;141(1):52–67. doi:10.1016/j.cell.2010.03.015.

[6] Pédelacq J-D, et al. *Engineering and characterization of a superfolder green fluorescent protein.* Nat Biotechnol. 2006;24:79–88. doi:10.1038/nbt1172. PDB: 2B3P.

[7] Abramson J, et al. *Accurate structure prediction of biomolecular interactions with AlphaFold 3.* Nature. 2024;630:493–500. doi:10.1038/s41586-024-07487-w.

[8] Yin J, et al. *Genetically encoded short peptide tag for versatile protein labeling by Sfp PPTase.* PNAS. 2005;102(44):15815–20. doi:10.1073/pnas.0507705102.

[9] Engelen W, et al. *Antibody-controlled actuation of DNA-based molecular circuits.* Nat Commun. 2017;8:14473. doi:10.1038/ncomms14473.

Eton College iGEM 2026 · Dry Lab · Aaran Nihalani · 2 July 2026 · Stack: Python (NumPy/SciPy/Matplotlib), AlphaFold3 Server, NUPACK 4